

Introduction & Hook

“When you move your head in VR, how does the system know where objects are relative to you?”

Everything in VR exists inside **mathematical coordinate systems**.
Without coordinate systems, rendering and tracking would be impossible.



PART 1: Coordinate Systems in VR

Basic Coordinate System

A coordinate system defines the position of objects using numbers.

In 3D space, we use:

- X-axis → left/right
- Y-axis → up/down
- Z-axis → forward/backward

Each object has coordinates like:

(x , y , z)

Example:

A cube at position:

(2, 3, 5)



PART 2: Types of Coordinate Systems

Local Coordinate System (Object Space)

Definition:

Local coordinates describe an object's position relative to itself.

Example:

- A door handle attached to a door
- The handle position is defined relative to the door

If the door moves:

→ The handle moves automatically.

Key point:

Local space is relative to the object's origin.

World Coordinate System

Definition:

World coordinates define an object's position relative to the entire virtual world.

Example:

- A building at (10, 0, 20) in the VR city
- A tree at (15, 0, 30)

World space:

- Is the global reference
- All objects ultimately exist in world space

Explain:

The engine converts local coordinates into world coordinates.

View (Camera) Coordinate System

Definition:

View coordinates describe object positions relative to the camera (user).

In VR:

- The camera = user's head
- As user moves → view coordinates change

Example:

If a cube is in front of you:

- View space Z value is positive

If you turn your head:

- View coordinates update instantly

Explain:

Before rendering, objects are transformed into view space.



PART 3: Transformations in VR

Types of Transformations

Transformations change object position or shape.

1. Translation (Movement)

Moves object from one location to another.

Example:

$(2, 3, 5) \rightarrow (5, 3, 5)$

Used when:

- Player walks
 - Object moves
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2. Rotation

Rotates object around X, Y, or Z axis.

Used when:

- User turns head
 - Object spins
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3. Scaling

Changes object size.

Used when:

- Zooming
 - Resizing UI elements
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Important Concept: Transformation Order

In VR rendering pipeline:

1. Local Space
2. World Space
3. View Space
4. Projection Space
5. Screen Space

Local → World → View → Projection → Screen

Real VR Example

Example:

In a VR classroom:

- A chair's local coordinates define its parts
- Its world coordinates place it inside the room
- View coordinates depend on where student is looking

If student walks:

- View matrix changes
- World stays same

This clarifies relative transformations.

Recap + Questions

Recap:

- Local space = relative to object
 - World space = global reference
 - View space = relative to camera
 - Transformations move objects between spaces
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Quick Oral Questions:

- What is difference between local and world coordinates?
- Why do we convert to view space before rendering?
- What happens if transformation order is wrong?